Carolin Schmidt, Ph.D.

ML researcher working on learning-based control for robotic and autonomous systems, with a focus on offline and hierarchical RL and structured policies that integrate classical control and optimization for scalability, robustness, and feasibility under operational constraints. Collaborations span Stanford, DTU, MIT, TUM, and Google DeepMind.

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Experience

10/2025– Present **Technical University of Munich | Postdoctoral Researcher / ML & Optimization Engineer**
- Developing a fast-and-slow planning architecture for control, in joint work with Stanford Autonomous Systems Lab; leading research direction as senior author.
- Leading first-author project on offline reinforcement learning for combinatorial action spaces, where the large-scale discrete structure of feasible actions breaks standard offline RL approaches.
- Delivered a structured RL solution for SAP, integrating learned policies with combinatorial optimization for feasible decisions at scale; led a two-person team and owned technical direction and solution design.
- Mentor Ph.D. and graduate researchers on project scoping, solution design, and research direction (10+ M.Sc. and 3 Ph.D. students supervised to date); co-lecture and organize ML and RL courses.

02/2025– 07/2025 **Technical University of Denmark | Postdoctoral Researcher**
- Led end-to-end development of the RL training and evaluation infrastructure for a large-scale climate-adaptation project; pipeline supported a case study presented to the Danish Parliament.

10/2023– 04/2024 **Stanford University | Visiting Researcher, Autonomous Systems Lab** (Prof. Marco Pavone)
- Developed OHIO, an offline RL framework for learning hierarchical, structured policies from logged data; demonstrated robustness to varying low-level controller configurations or changing environment constraints across robotic manipulation, goal-conditioned control, and network optimization tasks, outperforming end-to-end offline RL and imitation learning baselines.
- Designed a learning-based robo-taxi fleet controller that outperforms classical optimization and end-to-end learning on scalability and robustness at city-scale in high-fidelity simulation.

12/2021– 12/2024 **Technical University of Denmark | PhD Researcher, Machine Learning for Smart Mobility Lab**
- Built learning-based control frameworks for dynamic rebalancing and pricing of autonomous fleets across single-operator, multi-operator, and regulated-market settings.
- Designed and deployed a cloud-based arrival-time prediction API for autonomous shuttles in live pilots within the EU Horizon 2020 SHOW consortium.

Education

2021 – 2024 **Technical University of Denmark | Ph.D., Machine Learning and Optimization**
Thesis: *Prediction and Learning-based Control for Autonomous Mobility*
Advisors: Prof. Filipe Rodrigues, Prof. Francisco C. Pereira (DTU), Dr. Daniele Gammelli (Stanford)

2018 – 2021 **RWTH Aachen University | M.Sc. Electrical Power Engineering & Operations Research**
Grade 1.1, Honors (Springorum Commemorative Coin), Top 5 % (Dean's List)

2014 – 2018 **RWTH Aachen University | B.Sc. Electrical Power Engineering & Business Administration**
Grade 1.8, Top 5 % (Dean's List), Scholarship: RWTH Education Fund

Technical Skills

Machine Learning Offline, hierarchical, structured & multi-agent RL, imitation learning, neural combinatorial optimization, NN architectures, time series forecasting, computer vision (object tracking & segmentation), robustness & OOD evaluation

Optim. & Control Model predictive control, operations research, constrained and combinatorial optimization

Software Python, PyTorch (Lightning), Ray/RLlib, Gurobi, CPLEX, JAX, Docker, GCP, FastAPI, Hydra, DVC, Git, custom-built simulation & synthetic environments

Selected Publications

"Offline Hierarchical Reinforcement Learning via Inverse Optimization." *ICLR 2025.* Project page

"Robo-taxi Fleet Coordination at Scale via Reinforcement Learning." *IEEE Trans. on Control of Network Systems 2026.* github.com/StanfordASL/RL4AMOD

"Learning to Control Autonomous Fleets from Observation via Offline Reinforcement Learning." *European Control Conference 2024.*

"Synthetic Monitoring Environments for Reinforcement Learning." *Reinforcement Learning Conference 2026.* Package